

Nathaniel Sawant

Available for full-time positions starting January 2025

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EDUCATION

Northeastern University, Khoury College of Computer Sciences

Boston, MA

Bachelor of Science in Computer Science, Concentration in Artificial Intelligence

Sep. 2021 – Dec. 2024

GPA: 3.71/4.00

Leadership: Northeastern U. IEEE Chapter (President, *Fall 2023*): NUHOC (VP Publicity, *Summer 2024*)

Relevant Courses: Software Engineering, Database Management Systems, Machine Learning 1 & 2, Object-Oriented Design, Algorithms & Data Structures, C++ Programming, Natural Language Processing, Artificial Intelligence, Computer Systems, Mathematics of Data Models

Clubs: IEEE (Professional), NUHOC (Outdoors), NUPiC (Photography), Generate (Product Development)

PROFESSIONAL EXPERIENCE

Software Engineer

Jan. 2024 – Jun. 2024

Babel Street, Unstructured Team

Somerville, MA

- Automated report generation of 100+ project dependency licenses with Python and Maven plugins, saving hours
- Optimized Python training pipeline, using multi-segmented downloads to double speed and cut memory by 20%
- Setup Docker and Jenkins CI/CD pipeline for three Maven projects to perform integration and end-to-end testing
- Analyzed 2M+ items for categorical changes with Python & Wikidata ontology for Named Entity Recognition

Software Data Engineer

Jan. 2023 – Jun. 2023

Fidelity Investments, Advanced Data Analytics Team

Boston, MA

- Leveraged ETL process on 20+ sites to save metadata for 65,000+ items with Python and SQL for use by analysts
- Setup scrapers CI/CD using Jenkins and AWS Cloudwatch to store raw HTML in S3 and data in SQL databases
- Created data visualization dashboard with Angular to track usage of 120+ datasets to inform decision-making
- Improved productivity by integrating natural language search of code snippets within VSCode and JupyterLab

Research Assistant

Oct. 2021 – Aug. 2022

Northeastern University, Rehabilitation Games & Extended Reality Lab

Boston, MA

- Assisted in with multiplayer VR game with Unity and OpenXR to assess behavior of neurodivergent individuals
- Captured participants' eye-tracking data at 60HZ with HTC Vive Pro for analyzing changes in players' focus

EXTRACURRICULAR EXPERIENCE

Project Lead (DevDB, Snapper)

May. 2024 – Present

Generate Product Development at Northeastern University

Boston, MA

- Led and collaborated with 5 and 13 person Agile teams to manage project priorities, deadlines, & deliverables
- Defined system design & led development for internal database management tool using Node.js, AWS, & Postgres
- Analyzed 5 competitors & coordinated with client, design, tech leads to plan/develop 6 deliverables during semester

Software Engineer (Voxeti, Tubender)

Sep. 2023 – Apr. 2024

Generate Product Development at Northeastern University

Boston, MA

- Developed RESTful APIs with Golang, MongoDB, and JWT tokens for user registration and authentication
- Modified the open-source CuraEngine to estimate print time and material usage of 3D prints without side effects
- Implemented frontend components/pages in Electron and React-based desktop app, interacting with C++ backend

PROJECTS

Fish Species Computer Vision Classifier | *Python, PyTorch*

Sep. 2024 – Present

- Building image classification models to accurately predict almost 200 fish species from iNaturalist dataset
- Implementing training and evaluation pipelines for Convolutional Neural Network and Visual Transformers

Gear Rental Site | *TypeScript, Postgres, Prisma, Next.js, tRPC, NextAuth.js*

Apr. 2024

- Implemented gear rental site for 150+ member club with gear item checkout/return and SSO authentication
- Designed 6 pages of Figma front end with shared consistent design (typography, components, palette)

Reinforcement Learning Agent | *PyTorch, Unity, C#*

Mar. 2023

- Trained and tested reinforcement learning agent that seeks target in unknown continuous environment
- Built procedurally generated maze with Unity for training and testing of model over hundreds of epochs

TECHNICAL SKILLS

Languages: Python, C#, Golang, C/C++, SQL, Java/TypeScript, HTML/CSS

Frameworks: React, Next.js, Unity, JUnit, FastAPI, Express

Developer Tools: Git, Docker, AWS (S3, EC2, RDS), NoSQL (MongoDB), Relational Databases (Oracle, Postgres)

Libraries: pandas, NumPy, Matplotlib, scikit-learn